

EXAM

This is a 3 hours exam. Class slides and any handwritten material are allowed.
YOU MUST WRITE LEGIBLY.

I – TRUE, FALSE, UNCERTAIN? (1/2 of the points)

For each of those statements, say whether it is true, false, or uncertain and explain why. You should target around 10 to 15 lines per question. Copying the slides will not bring any points.

- 1 – It is optimal that unemployment rate is zero.
- 2 – In the Ramsey model, the transversality condition states that the level of capital should tend to zero.
- 3 – Labour-augmenting technical progress is a necessary condition for long run growth of income per capita.
- 4 – Intellectual property rights have ambiguous effects on growth and welfare.
- 5 – The speed of convergence is high in the AK model.
- 6 – Time is continuous and the interest rate is r . Consider a blue asset (value V_b) that pays b per unit of time and a yellow asset (value V_y) that pays y per unit of time. A green asset is an asset that is blue and that can turn yellow according to a Poisson distribution with intensity α . Using a discrete time approximation and taking the limit when the length of the period tends to zero, one can prove that

$$rV_g(t) = b + \alpha (V_b(t) - V_y(t)) + \dot{V}(t)$$

- 7 – Unemployment benefits increase unemployment because they shift upwards the Beveridge curve.
- 8 – With Nash-bargaining, the surplus of a match is efficiently shared.
- 9 – Hours worked per person cannot grow along a balanced growth path.
- 10 – In the real business cycle model, stabilizing output following a productivity shock is Pareto improving.

II – ENDOGENOUS GROWTH (1/2 of the points)

Time is continuous. We use the notation $X_t = X(t)$. Consider an economy in which population at time t is L_t and grows at the constant rate n , such that $\dot{L}_t = nL_t$. All agents have preferences given by

$$\int_0^\infty e^{-\rho t} \frac{C_t^{1-\theta} - 1}{1-\theta} dt,$$

where C is consumption defined over the final good of the economy. This good is produced as

$$Y_t = \left[\int_0^N y_t(i)^\beta di \right]^{\frac{1}{\beta}},$$

where $y(i)$ is intermediate good i and $0 < \beta < 1$. The production function of each intermediate good is

$$y_t(i) = \ell_t(i),$$

where $\ell(i)$ is labor allocated to good i . The wage is denoted w_t .

New goods are produced by allocating workers to the R&D process, with the production function

$$\dot{N}_t = \eta N_t^\phi \times L_{Rt},$$

where $\phi \leq 1$ and L_{Rt} is labor allocated to R&D. Labor market clearing requires

$$\underbrace{\int_0^N \ell_t(i) di}_{L_{Pt}} + L_{Rt} = L_t.$$

L_{Pt} is the amount of labor required for production.

Risk-neutral firms hire workers for R&D. A firm who discovers a new good becomes the monopoly supplier, with a perfectly and indefinitely enforced patent. There is free entry on the R&D market.

For the time being, let's consider the case $\phi = 1$ and $n = 0$.

1 – Write the instantaneous profit of the final good firm. Show that the demand for intermediate good i is

$$y_t(i) = p_t(i)^{\frac{1}{\beta-1}} Y_t.$$

2 – Show that profit maximization of each intermediate firm results in them setting price as a constant percentage markup over their marginal cost.

3 – Show then that demand for good i can be written as function of Y and w .

4 – Using the previous result and the production function of final good, determine the equilibrium wage as a function of N_t .

5 – Using the demand for good i , express L_{Pt} as a function of Y and N .

6 – Show that the profit of an intermediate firm can be written as a function of Y and w .

7 – We now turn to the R&D sector. Let Z_t denote the present discounted value of the profits of an intermediate good producer at time t . Next consider a firm undertaking R&D to create a new intermediary. Explain why does the free entry condition on the R&D market writes

$$Z_t \dot{N}_t - w_t L_{Rt} = 0.$$

8 – Derive an expression for Z as a function of N . (Hint: use the law of motion of N)

9 – Let us restrict to balanced growth path (BGP) and guess that N grows at a constant rate x (needed to be verified at some point). Using that guess and the expression for Z , derive the growth rate of Z along a BGP.

10 – Why does the the following equation holds? What does it mean? (r is the interest rate, π_t is the instantaneous profit of any intermediate firm at a symmetric equilibrium where all intermediate firms are taking the same decisions; this is the equilibrium that we consider here)

$$rZ_t = \pi_t + \dot{Z}_t.$$

11 – Using all the previous equations, derive an expression for Y as a function of r, x, N_0 and t . Express the growth rate of Y (and hence C) as a function of β and x . Denote this equation (1).

12 – Without doing the algebra, explain how do we get the following relation from the household intertemporal maximization problem:

$$\frac{\dot{C}_t}{C_t} = \frac{1}{\theta}(r - \rho). \quad (2)$$

How do you interpret this relation?

13 – Use (1) and (2) to obtain a relation (3) between x and r . Comment.

14 – We now need to determine x in equilibrium. Use the R&D technology and (3) to derive a equation between L_R and r . Call the relation ($\mathcal{A}$).

15 – Using the expression of L_P of question 5 and the one of Y in question 11, get an expression of L_P as a function of r and x . Using $L_R = L - L_P$, obtain a second relation ($\mathcal{B}$) between L_R and x .

16 – Compute the equilibrium level of L_R . Check that x is indeed constant and derive its equilibrium level. Compute also the equilibrium level of r .

17 – Compute the growth rate of the economy (which is the growth rate of Y or C). How does it depend on θ ? Why?

18 – How does the growth rate depend on L ? Why? Discuss the empirical relevance of that result.

19 – **We now want to characterize the balanced growth path equilibrium when $\phi < 1$ and $n > 0$.** Explain how to derive the new free entry condition

$$Z_t \eta N_t^\phi L_{Rt} - w_t L_{Rt} = 0.$$

20 – Derive an expression for Z_t as a function of N_t .

21 – Using $rZ_t = \pi_t + \dot{Z}_t$ and the expression of the profit, derive the growth rate of Y as a function of x .

22 – Derive the growth rate of L_{Pt} as a function of x .

23 – Derive the growth rate of Y assuming that the economy began on a balanced growth path at date 0.

24 – Does the growth rate of the economy depend on L ? Why? Discuss.